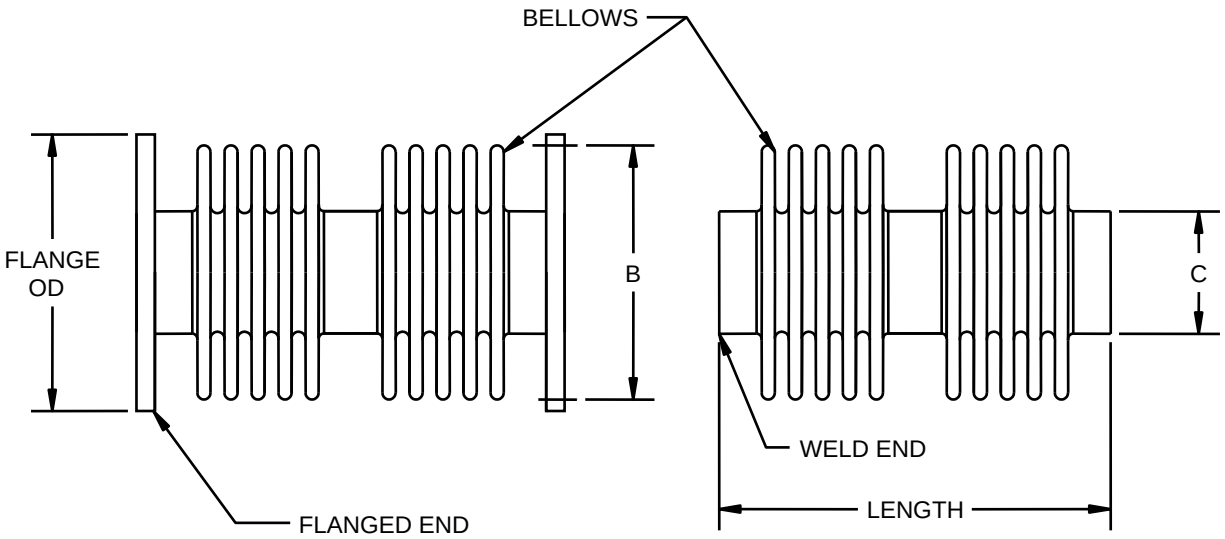
DETAIL SPECIFICATION SHEET

EXPANSION JOINTS, PIPE, METALLIC, DOUBLE BELLOWS, CORRUGATED, UNREINFORCED

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17813.



DN	C-NPS inch (mm)	Flange OD inch (mm) 1/	B (bolt circle diameter) inch (mm) 1/
50	2 (50.8)	6 (152.4)	4.75 (120.7)
65	2.5 (76.2)	7 (177.8)	5.5 (139.7)
80	3 (76.2)	7.5 (190.5)	6 (152.4)
90	3.5 (101.6)	8.5 (215.9)	7 (177.8)
100	4 (101.6)	9 (228.6)	7.5 (190.5)
125	5 (127)	10 (254.0)	8.5 (215.9)
150	6 (152.4)	11 (279.4)	9.5 (241.3)
200	8 (203.2)	13.5 (342.9)	11.75 (298.5)
250	10 (254)	16 (406.4)	14.25 (362.0)
300	12 (304.8)	19 (482.6)	17 (431.8)
350	14 (355.6)	21 (533.4)	18.75 (476.3)
400	16 (406.4)	23.5 (596.9)	21.25 (539.8)
450	18 (457.2)	25 (635)	22.75 (577.9)
500	20 (508)	27.5 (698.5)	25 (635)
550	22 (558.8)	-	-
600	24 (609.6)	32 (812.8)	29.5 (749.3)

1/ See ASME B16.5, flange dimensions,150 pound.

FIGURE 1. Expansion joint, double bellows, universal.

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NOTES:

1. Dimensions are in inches. Metric equivalents are given for information only.
2. The illustration is for identification purposes only and is not intended to restrict designs and shapes of fastening devices, which conform to the requirements and are within the envelope dimensions.

FIGURE 1. Expansion joint, double bellows, universal – Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Construction: A double expansion joint shall consist of two bellows joined by a common section of pipe. The common pipe section shall provide a means for anchoring the unit rigidly to prevent movement of the common section after installation.

The classification of the expansion joints covered under this specification sheet are as follows:

Type I - shall have corrugated bellows.

Class 2 - shall have two bellows separated by a section of pipe.

Class 2 expansion joints shall be with a base suitable as an intermediate anchor. (Note: An expansion joint that is used as an untied or tied universal should not have an anchor base.)

When specified (see MIL-DTL-17813), a drain connection shall be provided on the center pipe section of class 2 expansion joints and shall be threaded, flanged, or beveled for welding.

Materials and finishes.

Materials and finishes for the bellows and flanges shall be in accordance with MIL-DTL-17813.

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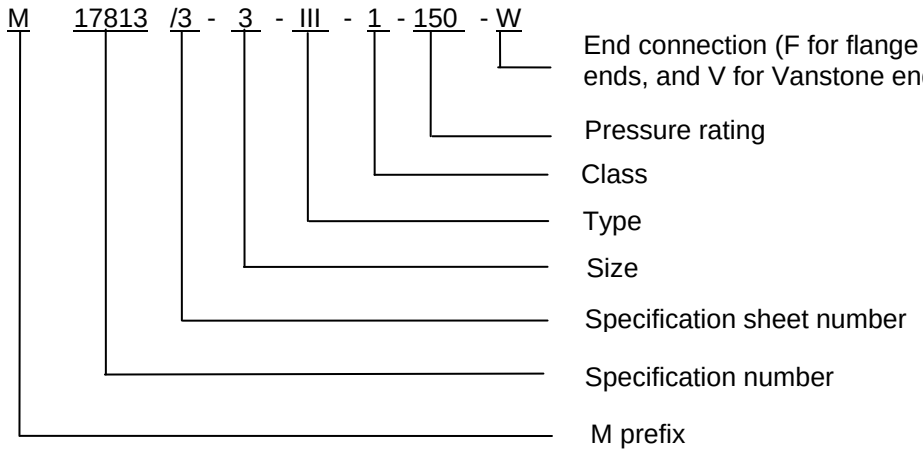
The Part or Identifying Number (PIN) consists of the letters M, the basic specification number, specification sheet number, size, type, class, pressure rating, and end connection.



PIN example: M17813/3-3-I-4-150-W

Referenced documents. In addition to MIL-DTL-17813, this document references the following:

ASME B16.5

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2009-101)

Review activities:
Army - CE
Navy – MC, SA
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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